BFW STEAM HEATER VER 1

Operator Training Simulator · E-101

Chulalongkorn University · Department of Chemical Engineering

ELECTRICAL LOAD LIST

DOCUMENT NO. : BFWH-V1-EE-LL-001

DISCIPLINE : Electrical

PROJECT : BFW Steam Heater Ver 1 · E-101 Operator Training Simulator

STATUS : Issued for teaching (IFT)

Electrical load list of the BFW steam heater unit E-101 with P-101A/B: motor rating from the simulator pump curve (37 kW, In 66.8 A), maximum demand in normal running 34.4 kW / 39.7 kVA, motor cable 25 mm² for an 80 m route (assumed), and the hazardous-area basis (unclassified).

Rev	Date	Description	Originator	Checker	Approver
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Numbers are taken from the BFW Steam Heater Ver 1 simulator model; values marked assumed / TBC / HOLD are not design data.

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1 Purpose and scope

This list gives every electrical consumer of the E-101 unit, the power it absorbs, its duty and the resulting maximum demand, so that the supply transformer, the motor control centre (MCC) and the UPS can be sized. It also records the rating of the BFW pump motors, the motor cable selection and the hazardous-area basis for equipment selection.

E-101 itself has no electrical load. The consumers are the BFW pump motors P-101A (duty) and P-101B (spare), the solenoid of the SIS shut-off valve XV-100, the DCS and SIS cabinets (which also loop-power the field transmitters), area lighting and maintenance socket outlets.

2 Basis and assumptions

Item	Value	Source
Pump curve	$Pd = 30 + 1000 - 0.03 \cdot Q^2$ kPa g	Simulator model (K.Ptk, K.H0, K.aP)
Design flow through the pump	110.7 m³/h (BFW 100 + recycle 10.7)	Simulator model, design snapshot
Discharge pressure at design	662.4 kPa g	PI-106 in the model
Maximum-load flow	125.6 m³/h	Operating windows (DS_OW fA100: FV-101A 100 %)
Pump shut-off pressure	1,030 kPa g	Model K.Ptk + K.H0
Pump efficiency	0.70	assumed (vendor curve TBC, EQ-50)
Motor margin on maximum absorbed power	1.15	assumed
Motor efficiency / power factor	0.93 / 0.86	assumed
Supply	400 V, 3-phase, 50 Hz; DOL starting	assumed
DOL starting current / pf at start	6 × In / 0.30	assumed
Cable route length MCC → pump	80 m	assumed (EQ-51)
Cable derating (ambient, grouping)	0.80	assumed
Voltage drop limits	running ≤ 5.0 %, starting ≤ 15.0 %	design criteria (assumed)
XV-100 solenoid	12 W, 24 V DC from the SIS UPS	assumed
DCS / SIS cabinets	1.5 / 0.8 kW on UPS	allowance, assumed
Lighting / small power	1.2 kW / 5.0 kW (diversity 0.2)	allowance, assumed

3 Load list

Diversity: 1 for continuous loads, 0 for the standby pump (one BFW pump runs at a time), 0.2 for socket outlets. Of the maximum demand, 2.31 kW is on UPS. During a pump change-over both motors run for a few minutes: add 29.9 kW for the transformer check, not for the continuous demand. AAV-101 (rev C automatic air vent) is thermostatic and self-acting: no electrical load.

Tag	Consumer	Supply	Rated kW	Absorbed kW	Load factor	η	pf	Duty	Consumed kW	kVA	Diversity	Demand kW	Demand kVA
P-101A	BFW pump motor (DOL), duty	MCC 400 V	37.0	27.78	0.75	0.93	0.86	Continuous	29.87	34.73	1.0	29.87	34.73
P-101B	BFW pump motor (DOL), spare — identical	MCC 400 V	37.0	27.78	0.75	0.93	0.86	Standby	29.87	34.73	0.0	0.00	0.00
XV-100 SOV	Solenoid of the SIS shut-off valve (energised to hold open, de-energise to trip)	SIS UPS 24 V DC	0.012	0.012	1.00	1.00	1.00	Continuous	0.012	0.012	1.0	0.01	0.01
AAV-101	Automatic air vent (rev C): thermostatic, self-acting — no electrical load	—	—	—	—	—	—	—	0.00	0.00	0.0	0.00	0.00
DCS cabinet	Controllers, I/O, loop power for the field transmitters	UPS 230 V	1.5	1.50	1.00	1.00	0.90	Continuous	1.50	1.67	1.0	1.50	1.67
SIS cabinet I-100	Logic solver, I/O, loop power for FT-101, TT-108, PT-100 trip inputs	UPS 230 V	0.800	0.800	1.00	1.00	0.90	Continuous	0.800	0.889	1.0	0.80	0.89
Lighting	Area and platform lighting around E-101 and P-101A/B	LDB 230 V	1.2	1.20	1.00	1.00	0.90	Continuous	1.20	1.33	1.0	1.20	1.33
Small power	Maintenance socket outlets	LDB 400 V	5.0	5.00	1.00	1.00	0.90	Intermittent	5.00	5.56	0.2	1.00	1.11
Total	**Normal running, one BFW pump**								68.25 connected			**34.4**	**39.7**

4 Motor rating of P-101A/B

Hydraulic power = Q·ΔP. At design Q = 110.7 m³/h, Pd = 662.4 kPa g: (110.7/3600) × (662.4 − 30) = **19.45 kW**; absorbed = 19.45 ÷ 0.7 = **27.78 kW**.

At maximum load Q = 125.6 m³/h, Pd = 556.8 kPa g: 18.38 kW hydraulic, 26.26 kW absorbed. With a constant efficiency Q·ΔP of this curve peaks at Q = √(H0/3a) = 105.4 m³/h: 27.89 kW absorbed, the largest value in the operating range.

Motor basis = 1.15 × 27.89 = 32.07 kW → next IEC standard size **37 kW**. Full-load current In = P/(√3·V·pf·η) = 37000/(1.732 × 400 × 0.86 × 0.93) = **66.8 A**; DOL starting current ≈ 401 A. Load factor at design 0.75. P-101B is identical.

Case	Q m³/h	Pd kPa g	ΔP kPa	Hydraulic kW	Absorbed kW
Design (FIC-101 100 + recycle)	110.7	662.4	632.4	19.45	27.78
Curve peak of Q·ΔP (constant η)	105.4	696.7	666.7	19.52	27.89
Maximum load (DS_OW fA100, FV-101A 100 %)	125.6	556.8	526.8	18.38	26.26

5 Motor cable sizing

Voltage drop ΔV % = √3·I·L·(R·cosφ + X·sinφ)/V × 100 with R at 20 °C × 1.2 for the operating temperature and X = 0.08 Ω/km (assumed). Selection for In = 66.8 A, 80 m: the smallest size whose derated rating ≥ In and whose running and starting drops are within the limits is **25 mm²** (derated 80.8 A; running 1.83 %, starting 4.69 %). Ratings and resistances are typical values for copper/PVC multicore cable in air (three loaded conductors), for teaching; the cable manufacturer's data govern.

Size mm²	Rating A (typical)	Derated A	R Ω/km 20 °C	ΔV running %	ΔV starting %	Result
1.5	18.5	14.8	12.1	28.98	61.51	fails: current, running ΔV, starting ΔV
2.5	25	20.0	7.41	17.78	38.08	fails: current, running ΔV, starting ΔV
4	34	27.2	4.61	11.10	24.09	fails: current, running ΔV, starting ΔV
6	43	34.4	3.08	7.45	16.45	fails: current, running ΔV, starting ΔV
10	60	48.0	1.83	4.46	10.20	fails: current
16	80	64.0	1.15	2.84	6.80	fails: current
25	101	80.8	0.727	1.83	4.69	**selected**
35	126	100.8	0.524	1.35	3.68	meets all limits
50	153	122.4	0.387	1.02	2.99	meets all limits
70	196	156.8	0.268	0.73	2.40	meets all limits
95	238	190.4	0.193	0.56	2.02	meets all limits
120	276	220.8	0.153	0.46	1.82	meets all limits
150	319	255.2	0.124	0.39	1.68	meets all limits
185	364	291.2	0.0991	0.33	1.55	meets all limits
240	430	344.0	0.0754	0.27	1.44	meets all limits

6 Hazardous-area classification

Unclassified (non-hazardous) area. All fluids of the unit are water: BFW, flash steam, condensate and the vent and trap losses of the same steam. None can form a flammable atmosphere, so no release from flanges, seals, PSV-100, the vent V-104 or the trap creates a hazardous zone. Motors, instruments and lighting may be industrial (non-Ex) equipment.

This changes if the unit stands within the zone of a neighbouring source (fuel gas, hydrocarbon pumps, tank vents): the plot-wide area classification drawing governs (TBC, plot layout outside the simulator). Oil carried in the flash steam is a fouling and condensate-quality issue (HAZOP R27), not a flammable release.

Outdoor motors, junction boxes and local control stations: IP55 (assumed), with canopy or drain plugs for tropical rain.

Personnel protection: shell at about 121 °C and hot discharges at the vent, the trap set and PSV-100; insulate or guard surfaces above about 60 °C (assumed touch limit) within reach and route discharges to a safe place (HAZOP R17).

7 Open engineering queries

No.	Subject	Query	Ref.	Status
EQ-50	P-101A/B pump curve and efficiency	The load list uses the model's quadratic pump curve (shut-off 1 030 kPa g) and a constant pump efficiency of 0.70 (assumed). With a constant efficiency the absorbed power peaks at 105 m³/h; on a real curve the efficiency falls past the best point and the power can keep rising to run-out (125.6 m³/h at FV-101A 100 %). Vendor curve with efficiency and NPSHr required to confirm the 37 kW motor.	BFWH-V1-EE-LL-001	Open
EQ-51	Motor cable route length	Cable sizing uses an 80 m route from the MCC to P-101A/B (assumed). Confirm the MCC location and the cable route on the plot plan; the 25 mm² selection holds up to the length at which the running drop reaches 5 % (see the calculator).	BFWH-V1-EE-LL-001	Open
EQ-52	Direct-on-line start of a 37 kW motor on the 400 V bus	DOL start at about 6 × In (≈ 400 A, assumed) is taken as acceptable. Confirm the supply transformer rating and short-circuit level allow it without a soft starter (voltage dip at the MCC bus, not only in the cable).	BFWH-V1-EE-LL-001	Open
EQ-53	Auto-start of P-101B (HAZOP R01)	R01 asks for an auto-start of P-101B on a P-101A trip or low PI-106 pressure. The MCC starter of P-101B needs a remote auto-start input from the DCS and a 'healthy / available' signal back; the starter scheme is HOLD until Instrumentation fixes the logic.	BFWH-V1-SA-HZP-001	Open

Notes

- Numbers come from the simulator model (pump curve K, design snapshot, Operating windows) through epc/tools/eelec_data.py; the Electrical tab computes the same values in the page and checks them.
- Values marked assumed are teaching assumptions, not vendor or site data.